

What is Lasso Regression?

Explain in detail from Zero to very advanced.
Make sure the explanation should be in pointers only, use very easy going and simple language such that even 15 year old student can understand. Make use of example for better understanding. Starting from the perspective of 15 year old student take the explanation to ML engineer level. Go from explanation to mathematical implementation then coding in python.



Below is a **step-by-step, pointer-only explanation** of **Lasso Regression**, starting from **zero level (15-year-old)** and smoothly going up to **ML engineer level**, with **examples** → **math** → **Python code**.

Language is **simple, friendly, and intuitive** throughout.

1 What problem are we solving? (Very basic)

- We want to **predict a number**
- Example:
 - Hours studied → Marks
 - House size → Price
- We use **Linear Regression** for this

2 Simple Linear Regression recap (Zero level)

- Formula:

- $y = mx + b$
 - Meaning:
 - x = input (hours studied)
 - y = output (marks)
 - m = how strongly x affects y
 - b = starting value
-

3 Multiple Linear Regression (Still simple)

- When there are **many inputs**
 - Example (Marks prediction):
 - Hours studied
 - Sleep hours
 - Attendance
 - Formula:
 - $y = w_1x_1 + w_2x_2 + w_3x_3 + b$
 - w_1, w_2, w_3 are **weights**
-

4 Problem with normal Linear Regression

- It **uses all features**, even useless ones
 - Example:
 - Marks depend on:
 - Hours studied ✓
 - Sleep ✓
 - Shoe size ✗ (useless)
 - Linear regression still gives **weight to shoe size**
 - Result:
 - Overfitting
 - Poor real-world performance
-

5 Idea behind Lasso Regression (15-year-old view)

- Lasso says:
 “If a feature is useless, make its weight **exactly zero**”
 - So:
 - Important features → keep
 - Useless features → remove
-

6 What does “Lasso” mean?

- LASSO = **Least Absolute Shrinkage and Selection Operator**
 - Big meaning, simple idea:
 - Shrink weights
 - Select features
-

7 Key superpower of Lasso Regression

- ✓ Performs **feature selection**
 - ✓ Removes useless features automatically
 - ✓ Makes model simple & clean
-

8 How Lasso does this (Intuition)

- Lasso **adds a penalty**
 - If weight is large → penalty increases
 - To reduce penalty:
 - Model pushes some weights to **exactly zero**
-

9 Loss function (Simple idea)

Normal Linear Regression minimizes:

- **MSE = average of (actual – predicted)²**

Lasso minimizes:

- **Error + Penalty**

10 Lasso Loss Function (Easy math)

$$\text{Loss} = \text{MSE} + \lambda \sum |w|$$

Where:

- MSE → prediction error
 - $|w|$ → absolute value of weights
 - λ (lambda) → penalty strength
-

1 1 What does lambda (λ) do?

- $\lambda = 0$
 - Same as Linear Regression
- Small λ
 - Slight shrinking
- Large λ
 - Many weights become 0

👉 λ controls how aggressive Lasso is

1 2 Example (Real-life style)

Dataset:

Predict Marks using:

- Hours studied
- Sleep hours
- Shoe size

After Lasso:

- Hours studied → weight = 12
- Sleep hours → weight = 5
- Shoe size → weight = 0

✅ Shoe size removed automatically

1 3 Why absolute value $|w|$ matters

- Lasso uses $|w|$, not w^2
- Absolute value creates sharp corners
- Sharp corners → weights hit **exactly zero**

✚ This is why Lasso removes features

1 4 Difference between Ridge & Lasso (Quick)

Feature	Ridge	Lasso
Penalty	w^2	
Makes weights small	✓	✓
Makes weights zero	✗	✓
Feature selection	✗	✓

1 5 Geometry intuition (Advanced but simple)

- Ridge → circular constraint
 - Lasso → diamond constraint
 - Diamond corners lie on axes
 - Solution hits axes → weight becomes zero
-

1 6 When should you use Lasso?

- When:
 - Too many features
 - Many features are irrelevant
 - You want simpler model
- Common in:

- Finance
 - Genomics
 - NLP (text features)
-

1 7 Mathematical formulation (ML engineer level)

Data:

- $X \in \mathbb{R}^{n \times p}$
- $y \in \mathbb{R}^n$

Objective:

$$\min_w \left(\frac{1}{2n} \|y - Xw\|^2 + \lambda \|w\|_1 \right)$$

Where:

- $\|w\|_1 = \text{sum of } |w|$
 - This is **L1 regularization**
-

1 8 Why optimization is harder than Ridge?

- $|w|$ is not differentiable at 0
 - Closed-form solution ❌
 - Uses:
 - Coordinate Descent
 - Sub-gradient methods
-

1 9 Feature scaling is VERY IMPORTANT

- Lasso is scale sensitive
- Always apply:
 - **StandardScaler**
- Otherwise:
 - Large-scale features dominate penalty

2 0 Python implementation (Step-by-step)

Step 1: Import libraries

python

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```
import numpy as np
import pandas as pd
from sklearn.linear_model import Lasso
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
```

Step 2: Sample dataset

python

 Copy code

```
X = np.array([
    [5, 7, 42],    # hours, sleep, shoe size
    [6, 6, 40],
    [7, 8, 41],
    [8, 7, 39],
    [9, 6, 43]
])

y = np.array([60, 65, 70, 75, 80])
```

Step 3: Feature Scaling

python

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```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Step 4: Apply Lasso

python

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```
lasso = Lasso(alpha=0.1)
lasso.fit(X_scaled, y)
```

Step 5: Check coefficients

python

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```
lasso.coef_
```

Example Output:

csharp

 Copy code

```
[10.2, 4.8, 0.0]
```

👉 Shoe size removed 🎉

2 1 Hyperparameter tuning (Advanced)

- Use Cross-Validation

python

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```
from sklearn.linear_model import LassoCV

lasso_cv = LassoCV(alphas=[0.01, 0.1, 1, 10])
lasso_cv.fit(X_scaled, y)
```

2 2 Bias–Variance tradeoff in Lasso

- Low $\lambda \rightarrow$ low bias, high variance
 - High $\lambda \rightarrow$ high bias, low variance
 - Goal $\rightarrow$ optimal λ
-

2 3 Pros & Cons (Engineer view)

✓ Pros

- Feature selection
- Simple model
- Prevents overfitting

✗ Cons

- Unstable with correlated features
 - Picks one feature, ignores others
-

2 4 Lasso vs ElasticNet

- ElasticNet = Lasso + Ridge
 - Better when features are correlated
-

2 5 Final one-line summary

Lasso Regression is Linear Regression with L1 penalty that automatically removes useless features by making their weights exactly zero.

If you want next:

- ✂ ElasticNet deep dive
- ✂ Lasso vs Ridge with graphs
- ✂ Interview questions
- ✂ Numerical derivation step-by-step

Just tell me 👍

Why absolute value $|w|$ matters

Lasso uses $|w|$, not w^2

Absolute value creates sharp corners

Sharp corners $\rightarrow$ weights hit exactly zero

📌 This is why Lasso removes features

Didn't understand above point



No problem 👍

Let me explain **VERY slowly and simply**, step-by-step, **no heavy math**, with **mental pictures**.

1 First: what is Lasso trying to do?

- Lasso wants:
 - Small error ✅
 - Some weights exactly = 0 ✅
- Weight = importance of a feature
- Weight = 0 $\rightarrow$ feature is **removed**

2 Two ways to punish weights

Ridge penalty

$$w^2$$

Lasso penalty

$$|w|$$

They look similar but behave **very differently**

3 Think of weight as a number line

lua

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<---- -3 -2 -1 0 1 2 3 ---->

4 How Ridge (w^2) behaves

- Square makes **everything smooth**
- Curve looks like a **bowl**

markdown

 Copy code

—
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- As $w \rightarrow 0$, slope becomes **gentle**
- Ridge says:

“Go close to 0, but no need to touch it exactly”

👉 Result:

- Weights become **small**
 - ❌ But **not zero**
-

5 How Lasso ($|w|$) behaves

- Absolute value looks like a **V shape**

arduino

 Copy code

\ /
\ /

$V \leftarrow$ sharp point at θ

- At $w = 0$:
 - Direction suddenly changes
 - Sharp corner
-

6 Why “sharp corner” matters (MOST IMPORTANT)

Imagine pushing a ball downhill.

Ridge (smooth bowl)

- Ball slows near 0
 - Keeps rolling
 - Stops **near** zero
 - **✗** Does NOT stick at zero
-

Lasso (V shape)

- Ball rolls down
- Hits the **sharp point**
- Gets **stuck there**

✓ That point = $w = 0$

7 Simple physical analogy

Ridge = smooth hill

- No flat stopping point
- You never stop exactly at center

Lasso = pit with a point

- Once you reach the center
 - You **stop exactly there**
-

8 Now combine this with optimization

Lasso minimizes:

$$\text{Error} + \lambda|w|$$

During training:

- Model keeps reducing weight
- When weight reaches 0
- Penalty is minimized
- Model says:

“Perfect, I’ll keep it zero”

9 Why this does NOT happen in Ridge

Ridge penalty:

$$w^2$$

- Gradient near zero is **very small**
 - No strong force to hit exactly zero
 - So:
 - Weight $\rightarrow 0.0003$
 - Not $\rightarrow 0$
-

10 Visual comparison (key memory trick)

Penalty	Shape	Result
w^2	Smooth bowl	Small weights
	w	

1 1 Feature removal connection

- Weight = 0

- Feature contributes nothing
- Feature is **deleted**

This is why:

Lasso automatically removes features

1 2 Ultra-short summary (exam ready)

- $|w|$ has a **sharp corner at zero**
 - Optimization gets stuck there
 - Weight becomes **exactly zero**
 - Feature is removed
-

If you want, I can explain this using:

- 📊 Graph drawing explanation
- 🧠 Geometry (diamond vs circle)
- 📊 Numerical example with values

Just tell me which one 🐼



In short tell he how is weight is set to zero



In very short: how Lasso sets weight to zero

- Lasso adds $|w|$ **penalty**
- $|w|$ has a **sharp corner at 0**
- While minimizing loss:
 - Weight keeps decreasing
 - When it reaches 0, it **gets stuck**
- So:

- $w = 0$
- Feature is removed

👉 Sharp corner at zero forces weight to become exactly zero



Didn't understand



No worries 😊

Let me say it in the **simplest possible way, one idea only.**

Imagine a dimmer switch (light control)

- Weight = brightness of light
 - Lasso has **one special rule**:
"If brightness becomes very small, turn the light **OFF completely.**"
-

What happens during training

- Model slowly **reduces weight**
 - Weight becomes:
 - $0.5 \rightarrow 0.2 \rightarrow 0.05 \rightarrow 0$
 - **At 0, Lasso says:**
👉 "Stop. Keep it OFF."
-

Why Ridge doesn't do this

- Ridge says:
"Make it dim, but don't turn it off"
- So weight becomes:

- 0.04, 0.02, 0.001 (never zero)
-

One-line takeaway (remember this)

Lasso is designed to turn small weights completely OFF (set them to 0).

If you want, I can explain it using **numbers only** or a **picture-style explanation** next 👍



Why is Ridge its w^2 and in LASSO its $|w|$?



This is a **very good question** 👍

I'll explain it **slowly, clearly, and logically**, starting from **intuition** → **reason** → **math**, in **simple language**.

1 First: what is regularization trying to do?

- Regularization means:
 - “Don't let weights become too large”
- Large weights = model depends too much on one feature
- This causes **overfitting**

So we add a **penalty on weights**

2 Why do we need a penalty at all?

- Loss = how wrong the prediction is
- If we only minimize loss:
 - Weights can become very large
- So we add:

3 Now the BIG question

Why choose w^2 or $|w|$ as penalty?

Because they behave differently near zero

4 Why Ridge uses w^2 (L2 penalty)

Intuition

- Squaring:
 - Makes big weights **very expensive**
 - Keeps small weights cheap

What w^2 does

- $2^2 = 4$
- $5^2 = 25$
- $10^2 = 100$

👉 Large weights are punished a lot

Effect of w^2 on learning

- Smooth curve
- Gradient exists everywhere
- Easy to optimize
- Shrinks weights **gradually**

📌 Result:

- Weights become **small**
 - ❌ Rarely become zero
-

Why Ridge wants this behavior

- When **all features** are useful
 - When features are **correlated**
 - Ridge spreads importance across features
-

5 Why LASSO uses $|w|$ (L1 penalty)

Intuition

- Absolute value:
 - Punishes weights **linearly**
 - Same pressure whether weight is small or big

What $|w|$ does

- $|1| = 1$
- $|5| = 5$
- $|10| = 10$

👉 No extra punishment for big values

Effect of $|w|$ on learning

- Sharp corner at zero
- Not smooth at zero
- Optimization **hits zero and stops**

📌 Result:

- Many weights become **exactly zero**
-

Why LASSO wants this behavior

- When **many features** are useless
 - When you want **automatic feature selection**
 - Simpler, interpretable model
-

6 One simple visual idea (no math)

Method	Goal
Ridge	"Use all features, but softly"
Lasso	"Use only important features"

7 Mathematical reason (simple)

Ridge penalty

$$\frac{d}{dw}(w^2) = 2w$$

- At $w \approx 0 \rightarrow \text{gradient} \approx 0$
- So weight **does not snap to zero**

Lasso penalty

$$\frac{d}{dw}(|w|) = \pm 1$$

- Constant push toward zero
- Strong pull even near zero

👉 Weight **hits zero**

8 Optimization perspective (engineer level)

Property	Ridge	Lasso
Differentiable	Yes	No (at 0)
Convex	Yes	Yes
Closed form	Yes	No
Feature selection	No	Yes

9 Final 1-line answer (EXAM / INTERVIEW)

Ridge uses w^2 because it smoothly shrinks all weights, while Lasso uses $|w|$ because its sharp behavior at zero forces some weights to become exactly zero, enabling feature selection.

If you want next, I can explain:

-  Geometry (circle vs diamond)
-  Numeric example
-  One-sentence memory trick

Just tell me 😊

